Smart Cliff Learning Solutions

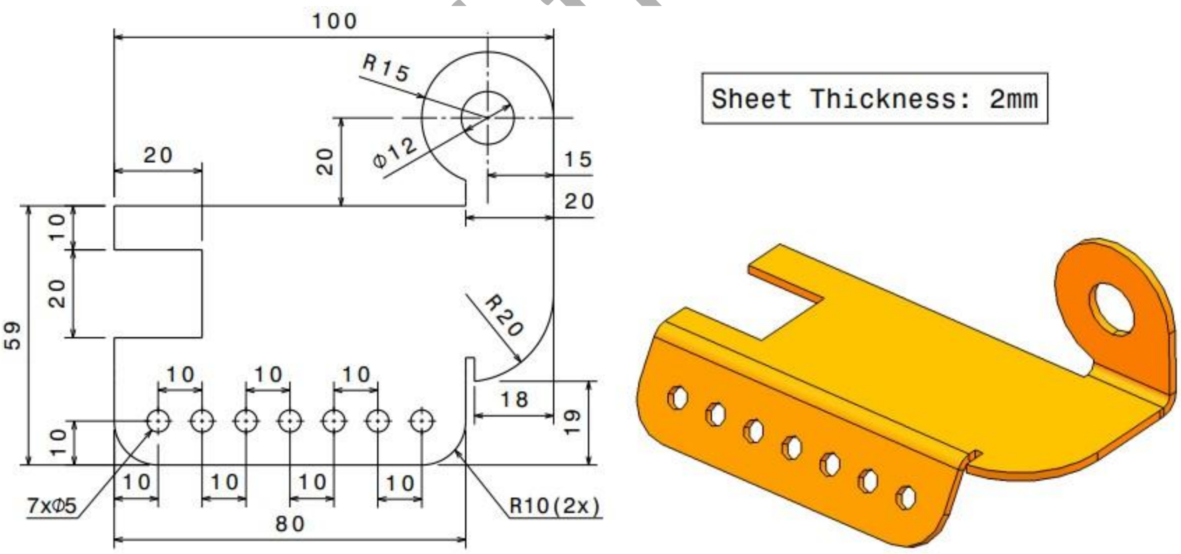
Technical Assessment

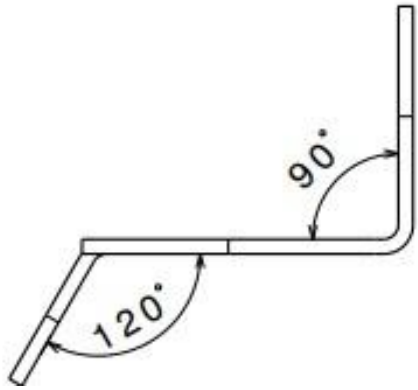
Client – JANATICS (2026)

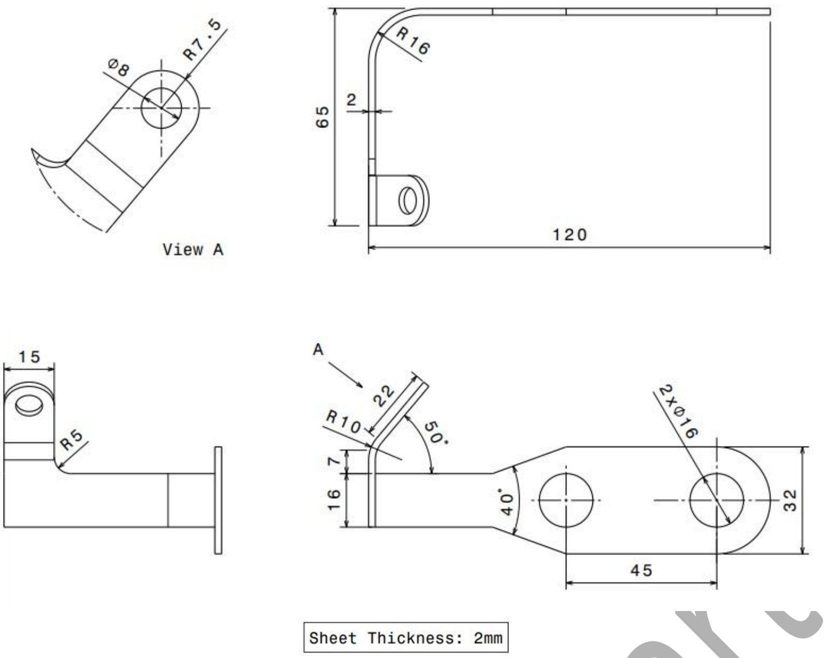
Module	**:**	**Tool (SolidWorks)**
Topic	**:**	**Sheetmetal Design**
Date	**:**	**30/04/2026**
Time	**:**	**2.00 – 5.00pm**
Duration	**:**	**3 Hrs.**
Total Marks	**:**	**50 Marks**

PART A: MCQ (10 x 1: 10 Marks)

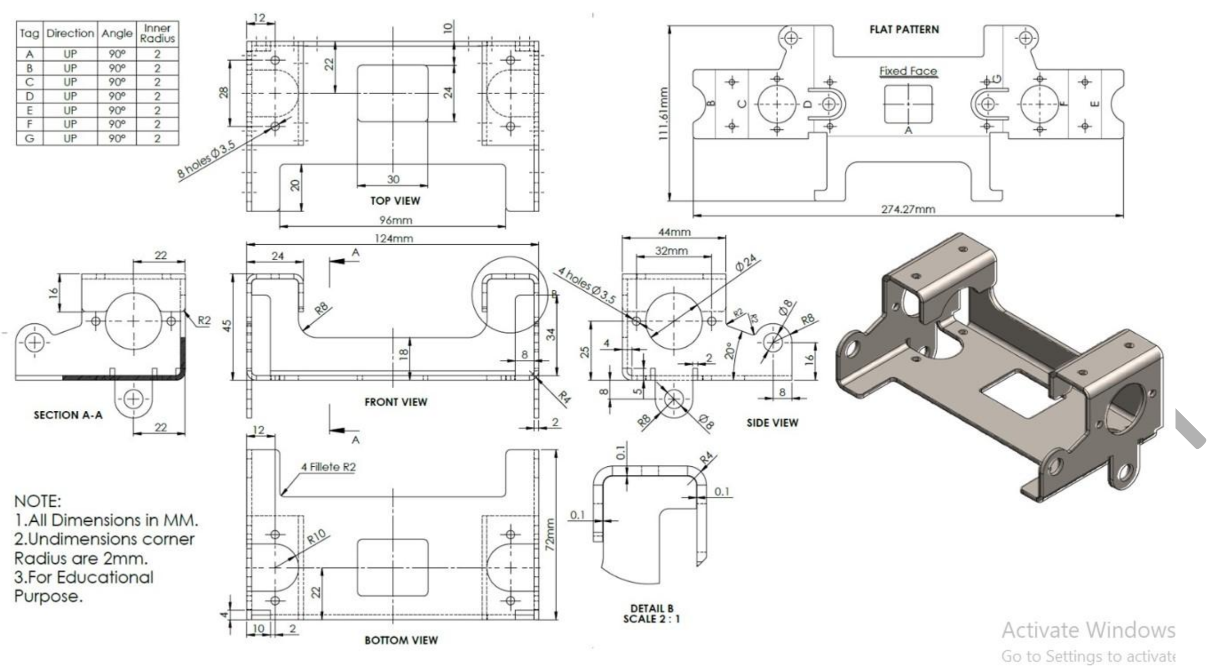
PART B: DESCRIPTIVE TYPE (40-MARKS)

Q.No		Level	Marks
1	**Design a given sheet metal component using SolidWorks, incorporating basic features such as a base flange, edge flanges, and necessary bends while maintaining a uniform sheet thickness; include a few holes or cutouts and ensure proper bend radius and sheet metal parameters so that the model is suitable for manufacturing, and generate the final outputs including the 3D model and flat pattern.** 	Medium	10

			
2	**Design a given sheet metal component using SolidWorks, incorporating basic features such as a base flange, edge flanges, and necessary bends while maintaining a uniform sheet thickness; include a few holes or cutouts and ensure proper bend radius and sheet metal parameters so that the model is suitable for manufacturing, and generate the final outputs including the 3D model and flat pattern.**	**Medium**	**10**

<table>
<tr><td></td><td>

</td><td></td><td></td></tr>
<tr><td>3</td><td>Design a given sheet metal model using SolidWorks, incorporating features such as a base tab, perpendicular flanges, and simple bends while maintaining uniform sheet thickness; include mounting holes or slots for assembly purposes and ensure appropriate bend radius and sheet metal parameters so that the part is manufacturable, and generate the final outputs including the 3D model and flat pattern.</td><td>Hard</td><td>20</td></tr>
</table>


Tag | Direction | Angle | Inner Radius
A | UP | 90° | 2
B | UP | 90° | 2
C | UP | 90° | 2
D | UP | 90° | 2
E | UP | 90° | 2
F | UP | 90° | 2
G | UP | 90° | 2
TOP VIEW
8 holes Ø3.5
96mm
124mm
FLAT PATTERN
Fixed Face
111.61mm
274.27mm
SECTION A-A
FRONT VIEW
SIDE VIEW
44mm
32mm
4 holes Ø3.5
Ø24
4 Fillete R2
BOTTOM VIEW
72mm
DETAIL B
SCALE 2 : 1
NOTE:
1.All Dimensions in MM.
2.Undimensions corner Radius are 2mm.
3.For Educational Purpose.
Activate Windows
Go to Settings to activate